

BioMarkHER2: A Pathologist-in-the-Loop System for Quantitative HER2 IHC Pre-Scoring Under Weak Supervision and CPU-Only Compute

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Abstract—Manual HER2 immunohistochemistry (IHC) scoring is subjective at its class boundaries, and the equivocal 2+ category triggers a second, costly confirmatory test. We present BioMarkHER2, an end-to-end AI-assisted measurement system for HER2 IHC intensity, built under two hard constraints: no real whole-slide-image data yet exists for the clinical site this project targets (84 slides, not digitized), and the development machine is CPU-only. The system comprises classical stain-based preprocessing, a weakly-supervised segmentation model trained without any pixel-level annotation, an evaluation framework combining conformal prediction, cross-group stain-variation analysis and offline ASCO/CAP mapping, and a local review-viewer web application through which a pathologist inspects the model’s output beside a classical control and records a confirmed score. The system’s central design principle – assistive, never autonomous – is formalized as a strict functional boundary between measurement and diagnosis and enforced as executable tests, not only as documentation: no API response or UI anywhere is permitted to carry a field named `score`, `her2_score`, `verdict` or `diagnosis`, checked on every test run. We report results across the full pipeline, an honest coverage accounting against the project’s original nine-item brief, a validity-threat analysis organized by the standard construct/internal/external/statistical taxonomy, and the items that remain genuinely blocked on data no engineering effort can substitute for, rather than on missing code.

Index Terms—digital pathology, HER2, immunohistochemistry, weak supervision, conformal prediction, human-in-the-

loop, clinical decision support

I. Introduction

HER2 status determines eligibility for HER2-targeted breast cancer therapy and is assigned by pathologists from immunohistochemistry (IHC) membrane staining on a 0/1+/2+/3+ scale, per the ASCO/CAP guideline [1]. The 2+ (“equivocal”) category cannot be resolved from IHC alone and triggers a second, more expensive confirmatory test (FISH/ISH); manual scoring at the 0-versus-1+ and 2+-versus-3+ boundaries is known to vary between observers. Against the backdrop of a broader, well-documented shift toward AI-assisted diagnosis in digital pathology [2], this motivates a quantitative, AI-assisted pre-scoring tool – one that measures staining and hands the measurement to a pathologist, rather than one that scores autonomously.

This paper reports on BioMarkHER2, a final-year academic system built with a specific medical college as its clinical reference point, under two constraints that shape nearly every engineering decision in the system:

- No real whole-slide-image data exists yet. The reference site’s 84 slides are not digitized. Every result in this paper is built and validated against an approved public stand-in dataset, `HER2_IHC_40X`, until real slides arrive.

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- The development machine is CPU-only. No GPU is available. Every model, batch size and epoch count in this system is sized to actually complete on a laptop CPU.

Rather than treat these as footnotes, we treat them as first-class design constraints and report their consequences explicitly throughout: which components are software-complete but only run at smoke scale, which objectives are fully exercised, and which are genuinely blocked on data rather than on missing engineering.

The system’s overriding design principle is stated once and then enforced everywhere:

This is a pre-scoring assistive tool that a pathologist reviews and confirms – never an autonomous final-scorer.

Concretely, this is not a policy documented in prose. No code path anywhere in the system is permitted to produce a field named `score`, `her2_score`, `verdict`, or `diagnosis`; a test suite scans the live application’s JSON output on every run and fails the build if one ever appears. A pathologist review can only be recorded through an explicit, human-initiated submit action. We describe this enforcement mechanism in Section VIII because we consider it as much a contribution as any individual model result: a framing constraint that is only ever documented, not tested, is a framing constraint a well-intentioned future edit can erode without anyone noticing.

This paper makes four contributions: (i) a complete, reproducible pipeline from raw IHC patch to a pathologist-facing measurement, built entirely under weak supervision and CPU-only compute; (ii) a head-to-head architecture comparison that resolves a diagnosed resolution failure on HER2 IHC’s most clinically consequential class; (iii) an evaluation framework – conformal prediction, stain-variation analysis, offline clinical-guideline mapping – that reports calibrated uncertainty and expert agreement rather than a bare accuracy number; and (iv) a systematic validity-threat analysis and honest coverage accounting against the project’s original brief, so that what remains blocked on data, rather than on engineering, is stated explicitly rather than left implicit.

II. Related Work

Computational pathology and HER2 IHC scoring automation. AI-assisted digital pathology has matured rapidly as a diagnostic and precision-oncology tool [2]. Automated HER2 scoring work within it typically targets patch- or slide-level classification of the 0/1+/2+/3+ label directly, including pyramid-sampling convolutional architectures for whole-slide HER2 scoring [3] and multi-stage CNN classifiers for HER2 category prediction [4]. Pixel-level segmentation of the staining pattern itself is comparatively unexplored, in part because pixel-level HER2 IHC annotations are not publicly available at scale – the gap this system’s weak-supervision pipeline (Section V) is built to work around, following a broader

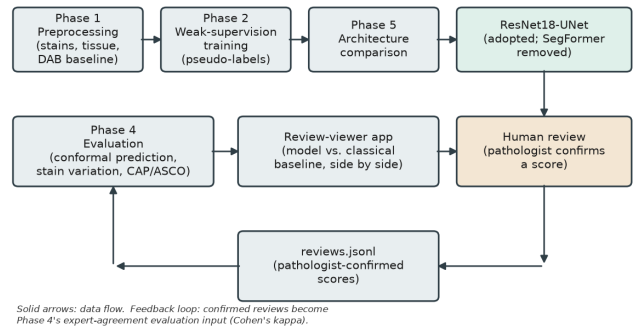


Fig. 1. System pipeline. Phase 1 preprocessing feeds Phase 2’s weak-supervision training; two architectures were compared head-to-head (Phase 5) and ResNet18-UNet was adopted. Phase 4 evaluates the model; the review-viewer application is the pathologist-facing surface, and confirmed reviews feed back into Phase 4’s expert-agreement evaluation.

precedent for deriving dense segmentation targets from patch-level classification labels in histopathology [5].

Human-in-the-loop clinical decision support. Systems in this space generally separate a measurement or triage step from the diagnostic decision itself; our review-viewer follows the same separation, with the “no score field” rule as its literal, tested boundary (Section VIII), rather than a convention left to interface design alone.

Stain variation. Colour and intensity variation across scanners and staining protocols is a well-known confound in computational pathology; recent generative approaches address it directly, e.g., with cycle-consistent adversarial networks for multi-domain stain normalization [6]. We take a complementary, non-generative approach – estimating a per-patch stain descriptor [11] and using it both to characterize cross-group variation (Section VII) and to reweight conformal calibration evidence (Section VII-B) – rather than learning a normalizing transform.

Conformal prediction for medical imaging. Conformal prediction [18] provides distribution-free, finite-sample-valid uncertainty for exactly this kind of assistive setting; recent work wraps segmentation output with morphological prediction sets [7] or quantifies region-level uncertainty for U-Net-based clinical tools [8], both as general-purpose, task-agnostic layers over an already-trained model. We build instead on a task-specific application of conformal prediction to binary, image-level HER2 status classification [9], generalizing it to pixel-level, five-class segmentation output (Section VII-B).

III. System Overview

Fig. 1 shows the system end to end. Phase 1 (Section IV) turns a raw RGB IHC patch into a tissue mask, a classical DAB-threshold intensity classification, and area percentages. Phase 2 (Section V) trains a segmentation model on pseudo-labels derived from Phase 1, since no pixel-level annotations exist. Phase 5 (Section VI) com-

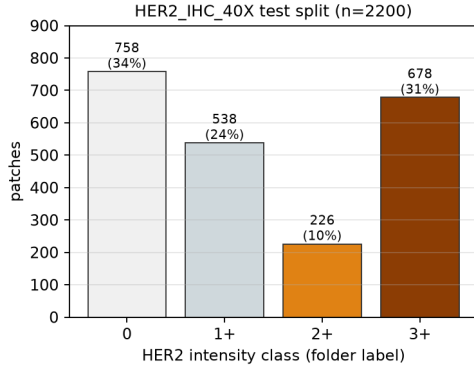


Fig. 2. HER2_IHC_40X test-split class distribution (folder label). Moderate (2+) – the class that decides reflex FISH testing – is the rarest, at 10% of patches.

compares two candidate architectures head-to-head and selects one. Phase 4 (Section VII) evaluates the selected model with conformal prediction, stain-variation analysis, and offline ASCO/CAP mapping. The review-viewer application (Section VIII) is the pathologist-facing surface that ties these together and is the only place a HER2 score is ever recorded, by a person, on explicit submission.

A. The Design Principle, Stated Formally

We state the system’s one enforced rule as a functional boundary, since a precise statement is what makes it testable rather than aspirational. Let $g_\theta : x \mapsto \hat{p}$ denote the entire measurement pipeline – preprocessing, the segmentation model, and evaluation – mapping an input image x to a vector of per-class area percentages and calibrated prediction sets $\hat{p}$. Let $h : (\hat{p}, \text{clinical judgment}) \mapsto \text{score} \in \{0, 1+, 2+, 3+\}$ denote the pathologist’s own act of scoring. The system’s one invariant is that no composition of functions reachable from a live API request implements any approximation of h : g_θ is never followed, inside the system, by a step that collapses $\hat{p}$ to a categorical score. Section VIII describes how this is checked, not merely declared, on every code change.

IV. Phase 1: Data and Preprocessing

A. Dataset

We use HER2_IHC_40X, $\approx 11,000$ (10,997) 1024×1024 RGB IHC patches at $40\times$ magnification, organised into four intensity folders with a slide-score filename prefix, as an approved substitute while the reference site’s slides remain undigitized. The dataset provides no pixel-level masks and no slide identifiers – both facts propagate into every later phase’s evaluation methodology and are restated where they matter, rather than only here. Fig. 2 shows the class distribution of the 2,200-patch test split used throughout this paper.

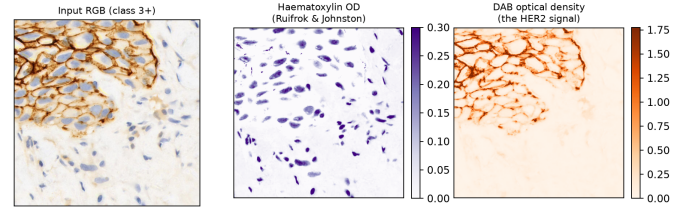


Fig. 3. Ruifrok-Johnston colour deconvolution on a real strongly-stained (3+) patch: input RGB, recovered haematoxylin channel, and recovered DAB channel.

B. Stain Deconvolution

HER2 IHC is dual-stained: haematoxylin (nuclear counterstain, blue) and DAB (the HER2 signal, brown). We separate the two by Ruifrok and Johnston colour deconvolution [10] in optical-density space. Fig. 3 shows this decomposition on a real, strongly-stained patch: the recovered DAB channel is the quantity every intensity class and every area percentage in this system is a function of. Stain normalization (Macenko [11], Reinhard [12]) is implemented but defaults to off, since HER2_IHC_40X is single-source and normalization would shift the exact optical densities the intensity thresholds are defined on without correcting any real variation.

C. Tissue Detection

The tissue mask is the denominator of every percentage the system reports, so its correctness matters more than its sophistication. An initial implementation used Otsu thresholding [13] on HSV saturation – standard for low-magnification whole-slide thumbnails. It failed on patch-level $40\times$ IHC: the chosen threshold tracked staining intensity rather than tissue presence, rising from ≈ 0.12 on faintly-stained patches to ≈ 0.43 on strongly-stained ones, silently excluding weakly-stained tissue from its own denominator. Fig. 4 demonstrates this on a real faintly-stained patch: the saturation-Otsu mask covers only 34.9% of the frame against 51.9% for an optical-density mask on the identical patch. We replaced saturation-Otsu with detection by mean optical density across RGB, which responds to any stain absorbing light rather than to which stain or how much. Saturation-based detection remains available and is the correct choice at low-magnification thumbnails, where the field is mostly glass; it is simply the wrong tool at patch-level $40\times$, where the field is already almost all tissue.

D. The Classical Baseline

A classical DAB-threshold classifier assigns each pixel one of five intensity classes (background, negative, weak, moderate, strong) at fixed optical-density cut points. This module plays three roles that matter to keep distinct: it is the pseudo-label generator Phase 2 trains on; it is the experimental control carried through every later

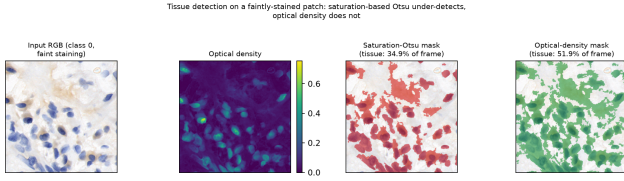


Fig. 4. Tissue detection on a real faintly-stained patch: saturation-Otsu under-detects (34.9%) relative to optical density (51.9%) on the identical frame.

evaluation, since a learned model that merely agrees with the rule it was trained on has demonstrated nothing; and it is the area-percentage engine reused by the review-viewer and every report.

E. Tiling

Large images are split into fixed-size tiles by strided subsampling (never averaging, which would blend stained and unstained pixels and shift the optical densities the classes are defined on), tracking each tile’s position in both the working and original image frame – bookkeeping a future whole-slide stitching stage will depend on.

V. Phase 2: Weakly-Supervised Segmentation

A. No Pixel Annotations, No Slide Identifiers

Two facts shape this phase entirely. First, there are no pixel-level pathologist annotations anywhere in this project; the dense segmentation targets are pseudo-labels – literally the Phase 1 classical thresholder’s output, cached to disk so they can be inspected and disputed rather than trusted blindly. A monotonic sanity check (mean stained area rising with the dataset’s own patch label, at two independently retiled magnifications) is the only validation available in the absence of real annotations. Second, HER2_IHC_40X carries no slide identifiers, so a true slide-level train/validation split – required so validation measures generalization rather than memorisation of one slide’s tissue – is impossible. The held-out set used throughout this paper is instead defined by an inferred filename origin token (1,904 patches); the validation set is a stratified random draw from the remainder and is explicitly labelled not leakage-free. Both caveats are written into every artifact a training run produces.

B. Model and Loss

The segmentation model is a ResNet18 [15] encoder with a U-Net [14] decoder, trained on cross-entropy, soft Dice, and focal loss [17], with dihedral-only augmentation (flips and 90° rotations, chosen because any colour or angle transformation would shift the DAB optical densities the classes are defined on). The loss composition is not incidental: Fig. 5 shows the pixel-level class imbalance in the fit set (moderate (2+) is 3.6% of pixels, against 80.8% for background and negative combined), which focal loss and, optionally, inverse-frequency weighting are specifically intended to counteract. Section VI explains why

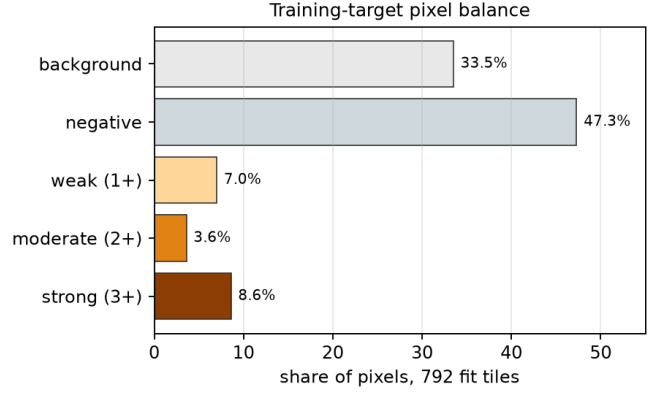


Fig. 5. Pixel-level class balance, 792 fit-set tiles. Moderate (2+) is the rarest class by pixel count.

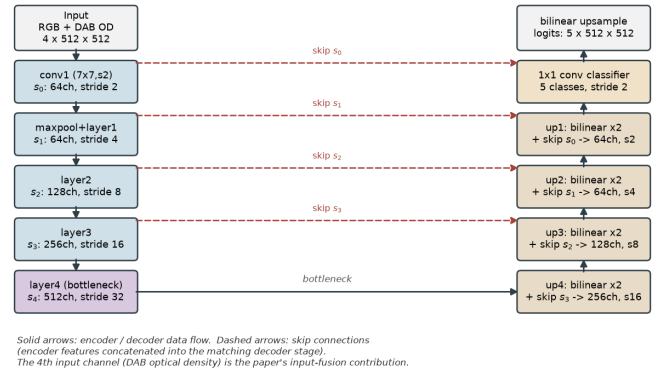


Fig. 6. ResNet18-UNet architecture, drawn from the model’s actual forward pass. The decoder stops at stride 2 (not stride 1), still twice SegFormer’s stride-4 output resolution, before a final non-learned bilinear upsample of the logits.

this architecture was adopted over an initial SegFormer baseline. Fig. 6 diagrams the adopted encoder-decoder, including the 4-channel (RGB + DAB) input and exactly which stride the decoder stops at.

VI. Phase 5: Architecture Comparison

An initial SegFormer-B0 [16] run diagnosed a specific failure: its decode head predicts at $H/4 \times W/4$ resolution before a single bilinear upsample, and at the magnification first tried, thin HER2 membrane rims (1–3 pixels wide) were sub-pixel at that resolution before ever reaching the loss. Retiling to native 40× resolution improved the weak (1+) class substantially (IoU 0.030 → 0.341) but left moderate (2+) essentially unlearned (IoU 0.0001, 253 of 21 million validation pixels). A second architecture – ResNet18 encoder, U-Net decoder, with skip connections carrying full-resolution encoder features into the decoder, plus the DAB optical-density channel fed as a fourth model input alongside RGB – was built and compared head-to-

TABLE I

Validation IoU, identical data/split/seed, 4 epochs, CPU-only.
Validation caveat: stratified random draw, not a held-out set (see Section V); used for architecture comparison, not as a generalization estimate.

Class	SegFormer-B0	ResNet18-UNet	Δ
Background	0.852	0.861	+0.009
Negative	0.815	0.839	+0.024
Weak (1+)	0.341	0.666	+0.325
Moderate (2+)	0.00006	0.589	+0.589
Strong (3+)	0.618	0.891	+0.273
Tissue mean IoU	0.443	0.746	+0.303
Pixel accuracy	0.862	0.908	+0.046

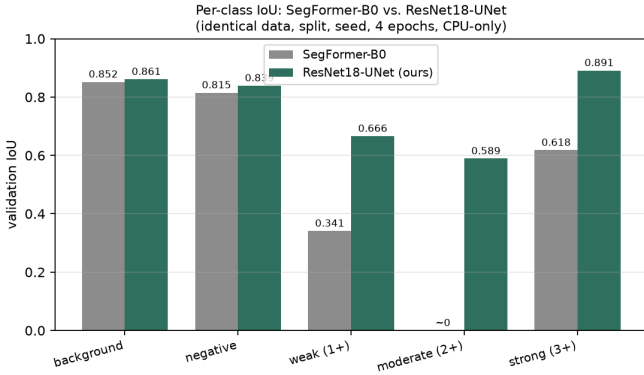


Fig. 7. Per-class validation IoU, SegFormer-B0 vs. ResNet18-UNet.

head against SegFormer, same data, split, seed and epoch count.

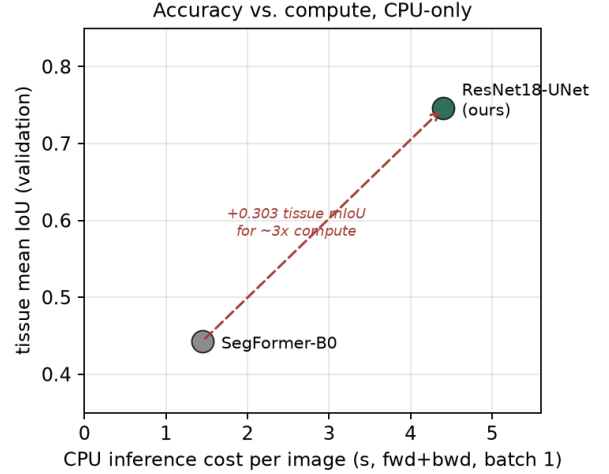
U-Net won on every class (Table I, Fig. 7), most dramatically on moderate (2+) – the class that decides reflex FISH testing. Decision: ResNet18-UNet was adopted as the sole architecture; SegFormer was removed from the codebase entirely, not kept as a fallback. Beyond the accuracy gain, U-Net’s plain Conv2d/BatchNorm/ReLU construction exports to ONNX and quantizes with fewer surprises than SegFormer’s attention-based decoder, and removing it dropped six now-unused dependencies. As Fig. 8 quantifies, at ≈ 4.4 s/image (forward + backward, CPU) against SegFormer’s ≈ 1.45 s, U-Net costs $\approx 3\times$ more per image – a real, stated price for the resolution it buys back, not an assumed free upgrade.

Fig. 9–11 show training behaviour, confusion structure, and qualitative predictions for the adopted model. Losses fell monotonically on train and validation with no divergence. Moderate (2+) remains the most-confused class (12.9% of its pixels called weak, 11.5% called strong) but is now a genuine, non-degenerate row in the confusion matrix rather than an empty one.

VII. Phase 4: Evaluation Framework

A. Stain Variation Across Provenance Groups

HER2_IHC_40X carries no slide identifiers, so cross-slide stain variation is measured across the dataset’s 8

Fig. 8. Accuracy vs. CPU compute, both architectures. The adopted model trades roughly $3\times$ compute for $+0.303$ tissue mean IoU.

coarser provenance groups (label $\times$ origin token) instead. A 6-dimensional stain descriptor (the haematoxylin and DAB rows of a per-patch Macenko-estimated stain matrix [11]) is compared within- and between-group over 25 sampled patches per group. The between-group mean descriptor distance (0.4230) exceeds the within-group mean spread (0.2802), i.e. `variation_exceeds_noise` is true (Fig. 12). This result must be read carefully: the 8 groups are confounded with HER2 score by construction (a 3+ group is, by definition, more heavily DAB-stained than a 0 group), so a stain descriptor built from haematoxylin and DAB directions will partly detect that confound rather than genuine cross-scanner or cross-institution variation. The result is genuine evidence the descriptor is sensitive enough to detect real variation of some kind; only real multi-institution data can say whether that kind is the cross-institution shift the weighting in Section VII-B is meant to correct for.

B. Stain-Shift-Weighted Conformal Prediction

We generalize binary, image-level inductive conformal prediction for HER2 status [9], [18] to pixel-level, five-class segmentation, using the hinge nonconformity score $s_c(x) = 1 - p_c(x)$ [19] and a per-class calibration quantile, and add stain-shift-weighted calibration [20]: each calibration score is weighted by a Gaussian kernel over the stain descriptor distance to the test patch, bandwidth set by the median pairwise calibration distance. Calibration and evaluation spend the reserved holdout set once, split into a calibration half and a test half that never touch the fit/validation data. Because no pathologist pixel annotations exist, calibration is against the same DAB-threshold pseudo-labels as training – a prediction set achieving nominal coverage here is well-calibrated against the rule the model was trained to imitate, not against clinical truth. This is the same generalize-then-weight

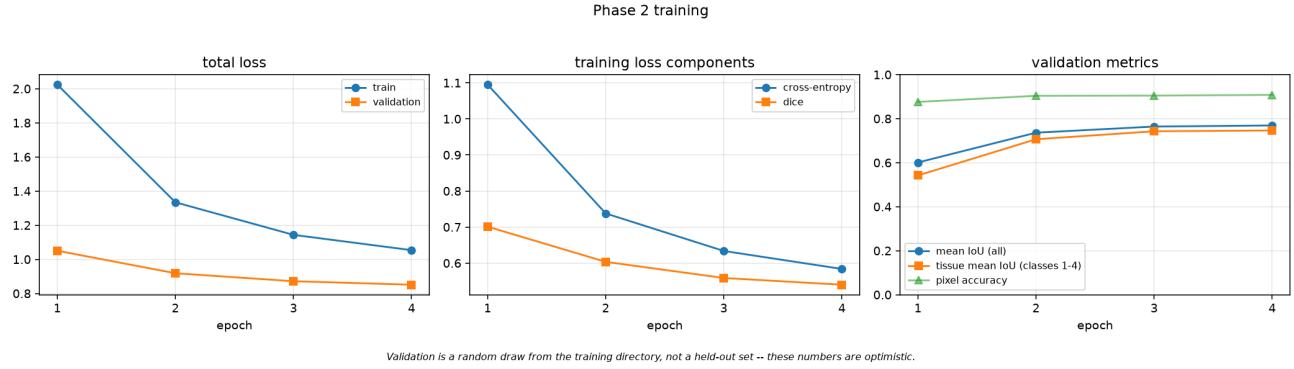


Fig. 9. Training loss, loss components, and validation metrics, adopted ResNet18-UNet model.

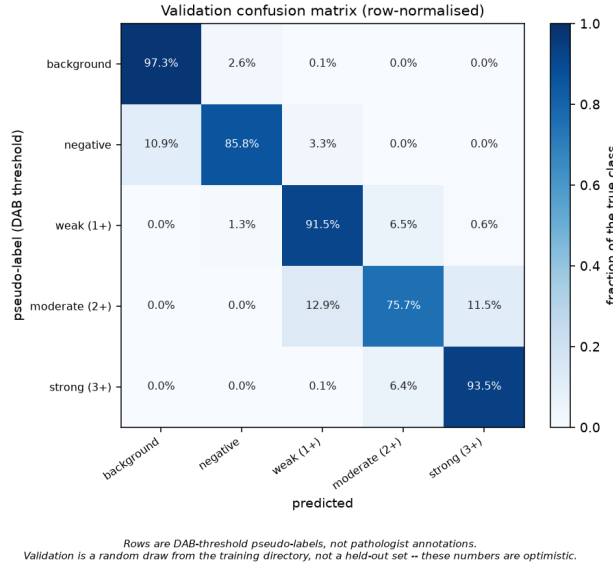


Fig. 10. Row-normalized validation confusion matrix, adopted model.

TABLE II

Pixel-level conformal prediction, smoke-scale test set (52 tiles, 10,595,420 tissue pixels), evaluated against the adopted checkpoint.

α	Unweighted			Weighted		
	Misc.	Amb.	Acc.	Misc.	Amb.	Acc.
0.05	0.044	0.110	0.971	0.047	0.106	0.969
0.10	0.089	0.072	0.950	0.093	0.072	0.949
0.20	0.183	0.142	0.953	0.189	0.151	0.955
0.50	0.489	0.480	0.983	0.491	0.482	0.983

construction developed independently in Section IV of the companion methods paper on this system’s conformal predictor; we summarize the empirical result here and refer the interested reader there for the full derivation.

Table II shows miscoverage tracking α closely (e.g. 0.044 at $\alpha = 0.05$) – the finite-sample coverage guarantee holding on real held-out data – and weighted vs. unweighted staying close at every row, which is the expected, honest result of running a cross-institution correction on

HER2_IHC_40X’s single-source data: there is no genuine shift here for the weighting to correct. The mechanism is validated separately on deliberately-constructed synthetic multi-source data in our test suite. This evaluation is smoke-scale (52 of $\approx 1,900$ available held-out test tiles); a full-scale run is software-ready but unrun, a multi-hour CPU job at this per-tile cost.

C. Offline ASCO/CAP Mapping and Expert Agreement

The model’s per-class area percentages are mapped to an ASCO/CAP 2018 [1] category by area-percentage thresholds analogous to the guideline’s tumour-cell-percentage rule: 3+ if strong-staining area exceeds 10%; else 2+ if moderate-or-above area is at least 10%; else 1+ if weak-or-above area exceeds 10%; else 0. This mapping is deliberately never imported into the review-viewer application (Section VIII) – exposing a cap_score field in the live API would be the system’s one enforced rule, broken under a different field name; formally, it would implement the very h that Section III-A forbids inside g_θ . It exists only as an offline script a person runs and reads. Agreement with expert ground truth is computed as exact agreement, within-one-category agreement, and quadratic-weighted Cohen’s kappa [21] against reviews.jsonl, the file the review-viewer’s /api/review endpoint appends a record to every time a pathologist confirms a score. As of writing, that file has zero real entries, because no pathologist has yet used the viewer – the pipeline is complete and tested against synthetic review data, and activates automatically the day a real review is recorded. We state this plainly rather than substituting a synthetic number for a clinical one. A further, permanent caveat: ASCO/CAP scoring is defined on percentage of tumour cells; every measurement in this system is percentage of tissue area, which includes stroma, and the two diverge whenever stroma content varies – which it does substantially here (mean tissue fraction 0.45 in class 0 to 0.79 in class 3+).

Phase 2 predictions vs DAB-threshold targets (validation patches)

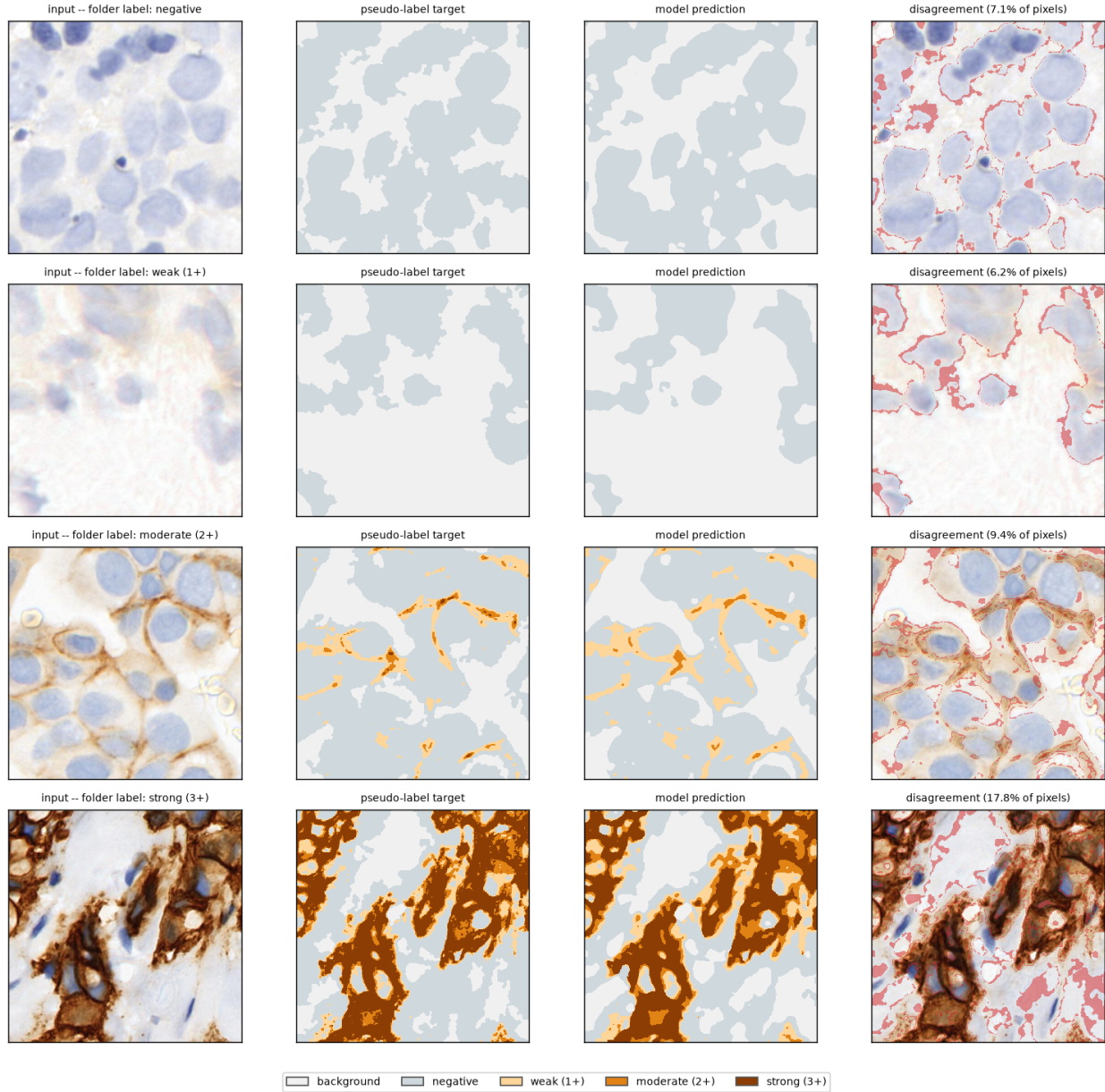


Fig. 11. Input, pseudo-label target, prediction and disagreement, one validation patch per intensity class, adopted model.

VIII. The Review-Viewer Application

A. Design

The review-viewer is a local, dependency-free web application (`http.server.ThreadingHTTPServer`; no framework, no CDN, nothing fetched from the internet at runtime), because the machines this system is meant to be demonstrated on – hospital and college computers – are exactly where an internet dependency or a heavyweight framework is a failure mode. It binds to 127.0.0.1 by default and warns loudly if pointed elsewhere: displaying medical images and recording clinical opinions with no authentication and no

transport security makes this a local demonstration and review aid, not a deployable clinical service, by design.

B. Workflow

Fig. 13 traces one review end to end, from opening the viewer to an optional PDF export. A pathologist opens the viewer, selects a sample patch or uploads an image, and receives four panels – original, detected tissue, model intensity map, classical baseline – and one area-percentage table, model and baseline always shown side by side, never one without the other: because the model trains on the baseline’s own output, agreement between

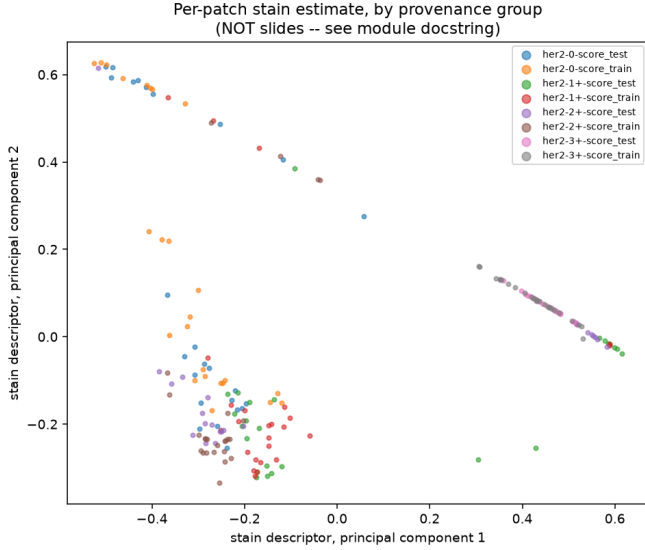


Fig. 12. Between- vs. within-provenance-group stain descriptor distance, HER2_IHC_40X.

the two is agreement with the rule the model imitates, not evidence of clinical accuracy, and hiding that comparison behind an optional toggle would let a viewer see only the flattering number. The moderate (2+) row is flagged inline in the table itself (an amber warning), not only in a footnote, because a caveat at the page bottom is a caveat a screenshot crops out. The pathologist then completes an explicit review form – a reviewer identity plus a chosen score, with “cannot assess from this field” offered as a first-class option – and only that submission writes a record to reviews.jsonl. A PDF export renders the identical payload the live API returns, carrying the same never-a-score guarantee, verified directly in the test suite.

C. The One Rule, Enforced as Code

The system’s central design constraint – assistive, never autonomous, and formally $h \notin \text{range reachable from } g_\theta$ (Section III-A) – is checked, not just stated: the live JSON API is scanned on every test run for the literal keys score, her2_score, verdict, and diagnosis, and the build fails if any appear anywhere in the response. The review UI’s submit action is likewise tested to require an explicit form submission with a reviewer identity; no code path can record a review automatically. We treat these as load-bearing specification, in the same category as ordinary correctness tests (tiled predictions cover the whole image; splits contain no leaked patches; weights resolve to the expected length), because they are exactly the properties a well-intentioned future UI or API change could erode without anyone noticing until a pathologist is looking at a silently overconfident screen.

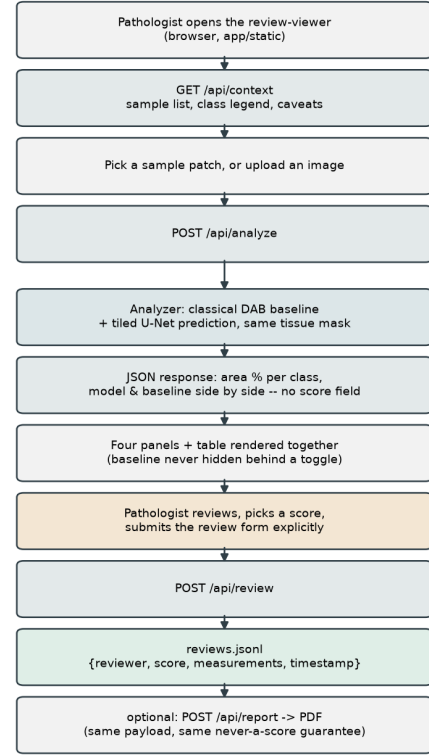


Fig. 13. Review-viewer request flow, one review end to end. A HER2 score enters the system only at the explicit-submit step; nothing upstream of it can create one.

D. Packaging

The application is containerized (Dockerfile plus docker-compose.yml), publishing on 127.0.0.1 on the host only, matching the server’s own no-broader-exposure stance. PyTorch is installed explicitly from the CPU wheel index rather than plain PyPI, since a naive pip install -r requirements.txt-only image would silently lack it. Docker packaging is written and unit-verified but has not been run to completion end to end on real hardware, as no Docker installation was available on the original development machine; we report this as an open item rather than a completed one.

IX. System-Level Results

Table III summarizes the honest state of every item in the project’s original brief: 6 of 9 are done, 1 (slide-level heatmaps) is genuinely partial pending whole-slide stitching, and 2 (real-slide digitization; validation against real pathologist annotation) are blocked on data no engineering effort in this system can substitute for – they are ready to activate the moment that data exists, not unstarted. The system’s test suite – 283 tests at time of writing, spanning every phase above – is treated as the project’s actual specification: a real category of these tests encodes the framing constraints of Section VIII as

TABLE III
Coverage against the project’s original nine-item brief.

Scope item	Status
Digitize IHC slides into WSI	Blocked on data
Tissue detection	Done
Stain normalization + deconvolution	Done
Patch extraction, patch-level analysis	Done
Intensity classification (weak/mod./strong)	Done, 1 class weak
Tissue area % per intensity category	Done
Heatmaps + slide-level summary	Partial
Validation vs. pathologist annotation	Built, 0 real data
Integration as pre-scoring workflow tool	Done

literal assertions, not only ordinary correctness checks, specifically so a future change cannot silently erode them.

X. Discussion: Honesty as an Engineering Principle

Two properties of this system are, we argue, as important as any single accuracy number. First, every reported number carries the conditions it was measured under, written into the artifact itself, not only into this paper: validation metrics are marked not-leakage-free at the point they are computed; conformal coverage results carry a pseudo-label calibration caveat; CAP-mapped agreement carries a tissue-area-not-tumour-cell-area caveat. A number copied out of this system’s output directory arrives with its caveat attached, because the caveat is written into the same file as the number. Second, framing constraints are tested, not merely documented (Section VIII): the distinction between “this tool measures” and “this tool scores” is the entire clinical safety argument for building it this way, and encoding that distinction as an executable test is what keeps a future, well-intentioned change from crossing it by accident.

XI. Threats to Validity

We organize the system’s limitations by the standard construct/internal/external/statistical validity taxonomy, since it makes explicit which threats are resolvable by more engineering and which are not.

Construct validity. Every accuracy, coverage, and agreement number this system reports is a measurement of agreement with a classical DAB-threshold rule (Section V) or, in the case of CAP agreement, with the dataset’s own folder label – not with independent pathologist ground truth, which does not yet exist for this project in pixel or, for CAP agreement, case form. Similarly, ASCO/CAP percentages are defined on tumour-cell area; this system reports tissue area, and the two diverge whenever stroma content varies.

Internal validity. The held-out set is inferred from filename structure on a dataset with no slide identifiers, not a verified slide-level partition – validation and holdout

performance may in part reflect generalization across patches of the same slide rather than across slides.

External validity. Every result in this paper is validated against HER2_IHC_40X, a public single-source substitute, not the reference site’s 84 (still undigitized) slides; the stain-shift-weighted conformal predictor has no genuine cross-institution shift on this dataset to demonstrate a benefit against and is validated on synthetic shift only; and Docker packaging is written and unit-verified but not run to completion end to end on real hardware.

Statistical conclusion validity. The conformal evaluation is real, computed against the actual deployed checkpoint, but over 52 of $\approx 1,900$ available held-out test tiles – a full-scale run is software-ready but unrun, a multi-hour CPU job at this per-tile cost. No result in this paper is averaged over multiple seeds, owing to the CPU-only compute budget.

Blocked on data, not on engineering. Two items in Table III – real whole-slide-image data and real pathologist reviews (reviews.jsonl has zero real entries) – are threats no further engineering in this system can resolve; both pipelines are complete and tested and activate automatically the moment the corresponding data exists.

The remaining open question. Moderate (2+) remains the weakest stained class (IoU 0.589 against 0.67–0.89 for the others) and is exactly the category that decides reflex FISH testing; it is resolved as learnable, not as solved.

XII. Broader Impact and Ethical Considerations

The system’s central engineering decision – that g_θ (Section III-A) never implements h – is also its central ethical safeguard: the risk we designed most deliberately against is automation bias, where a sufficiently accurate-looking assistive output is treated as a diagnosis by a time-pressured reviewer. Making the “measures, does not score” boundary a tested property of the API, not only of the interface copy, is intended to make that boundary durable across future changes made by people who were not in the room for this design decision. Conformal prediction reinforces the same goal at the statistical level: a wide prediction set at a clinically relevant significance level is a legitimate, intended output – the system declining to be confident – not a failure to be tuned away.

Beyond automation bias, two considerations bear on responsible deployment. Representativeness: HER2_IHC_40X’s demographic and institutional provenance is not documented at the level clinical deployment would require, and no result in this paper should be read as evidence of population-level clinical validity. Equity of access: every component in this pipeline runs on CPU-only commodity hardware by design (Section VI), which we view as directly relevant to whether AI-assisted pathology tooling remains reachable for resource-constrained departments rather than only for institutions with GPU infrastructure. Finally, the reviewer’s explicit no-authentication, no-transport-security,

localhost-only design (Section VIII) is a deliberate scope limitation, not an oversight: it is a demonstration and review aid for a controlled setting, and any path toward broader clinical deployment would require an authentication, auditing, and data-governance layer this paper does not claim to provide.

XIII. Conclusion and Future Work

BioMarkHER2 demonstrates that a defensible, pathologist-in-the-loop HER2 IHC pre-scoring system can be built end to end – preprocessing, weakly supervised segmentation, calibrated uncertainty, offline clinical-guideline mapping, and a tested, demonstrable application – entirely under weak supervision and CPU-only compute, with every result reported alongside the conditions it was measured under and the validity threats organized systematically rather than left implicit. Immediate future work is: whole-slide tiling and stitching against synthetic mosaics, buildable now without waiting for real WSIs; a full-scale conformal evaluation over the entire reserved holdout set; multi-seed replication of the architecture comparison; a per-cell membrane-completeness reformulation of the moderate (2+) class, closer to how ASCO/CAP itself defines the score; and, as real data arrives, replacing pseudo-label calibration with pathologist pixel annotation and exercising the stain-shift-weighted conformal predictor on genuine multi-institution variation.

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